

DEVIXO SOLUTIONS

Data Science Internship — Task 1

Exploratory Data Analysis & Statistical Analysis Report

Pakistan Air Quality Analysis (2015 – 2025)

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Internship Domain	Data Science
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GitHub Repository: https://github.com/MalaikaJabeen/Devixo_Solutions_Data_Science_Internship_projects/tree/main
Dataset: pakistan_air_quality_monthly_2015_2025.csv | 1,320 rows × 21 columns

Objective

The objective of this task is to perform a complete Exploratory Data Analysis (EDA) and statistical analysis on a self-selected dataset of 1,000+ rows as the first deliverable of the Data Science Internship at Devixo Solutions. The chosen dataset tracks monthly air quality (PM2.5 concentration and US AQI) across 10 major Pakistani cities from 2015 to 2025. The analysis aims to clean and validate the data, examine feature relationships and correlations, visualize temporal and geographic pollution patterns, and translate the statistical findings into actionable business insights and recommendations.

Dataset Description

The dataset contains 1,320 monthly observations (10 cities × 12 months × 11 years, 2015–2025) across 21 usable columns after cleaning. Each record includes the city, province, region, geographic coordinates, population, PM2.5 concentration ($\mu\text{g}/\text{m}^3$), US AQI value and category, the ratio of PM2.5 to the WHO guideline, and contextual flags for smog season, crop-burning season, monsoon season, and COVID-19 lockdown periods. The data spans 10 major cities: Faisalabad, Gujranwala, Hyderabad, Islamabad, Karachi, Lahore, Multan, Peshawar, Quetta, and Rawalpindi, covering all five provinces/territories of Pakistan.

Dataset Preview — `df.head(8)`

year	month_name	city	province	pm25_ugm3	aqi_us	aqi_category	season
2015	January	Faisalabad	Punjab	266.3	317	Hazardous	Winter (Smog Season)
2015	February	Faisalabad	Punjab	237.7	287	Very Unhealthy	Winter (Smog Season)
2015	March	Faisalabad	Punjab	145.0	197	Unhealthy	Spring
2015	April	Faisalabad	Punjab	74.7	161	Unhealthy	Spring
2015	May	Faisalabad	Punjab	59.6	153	Unhealthy	Summer/Monsoon
2015	June	Faisalabad	Punjab	55.0	149	Unhealthy for Sensitive Gr	Summer/Monsoon
2015	July	Faisalabad	Punjab	63.9	155	Unhealthy	Summer/Monsoon
2015	August	Faisalabad	Punjab	80.4	164	Unhealthy	Summer/Monsoon

Figure — First 8 rows of the cleaned dataset (`df.head()`).

Dataset Source

The data is compiled from the IQAir World Air Quality Report (2018–2025), IQAir city monitoring pages, a PMC research compilation, and the WHO Air Quality Life Index (AQLI) 2025 report. Records prior to the monitoring era are marked with a `data_quality` flag of "Estimated (pre-monitoring era)", while more recent years are marked "Verified", which was factored into the interpretation of long-term trends.

Technologies Used

- Python 3 — core analysis language
- Pandas & NumPy — data loading, cleaning, and aggregation
- Matplotlib & Seaborn — statistical visualization
- Jupyter Notebook — interactive analysis environment
- Microsoft Word / PDF tooling — this report's generation

Data Cleaning Process

The raw dataset (1,320 rows × 23 columns) was first inspected using `df.shape`, `df.info()`, `df.dtypes` and `df.isnull().sum()` to understand its structure. Two columns — `pakistan_national_avg_pm25` and `pakistan_global_rank` — were found to be populated for only 960 of 1,320 rows (360 missing values each, corresponding to the pre-monitoring era before 2018) and were dropped since they duplicated information better captured by the `per-city pm25_ugm3` and `aqi_us` columns. After this, `df.isnull().sum()` confirmed zero missing values across all 21 remaining columns. A full duplicate-row check (`df.duplicated().sum()`) returned 0, confirming the dataset is fully deduplicated. Categorical columns (`city`, `province`, `region`, `season`, `aqi_category`) were validated with `.unique()` and `.value_counts()` to confirm consistent, correctly-spelled category labels and a balanced panel of exactly 132 records per city.

```
1 # Finding null values in columns
2 df.isnull().sum()
3
4 # Checking for duplicate rows
5 df.duplicated().sum()          # 0 duplicates found
6
7 # Removing unnecessary / mostly-empty columns
8 df = df.drop(
9     columns=['pakistan_national_avg_pm25', 'pakistan_global_rank'],
10    errors='ignore'
11 )
12 df.isnull().sum()             # confirms 0 missing values remain
```

Code — Null-value check, duplicate check, and column removal.

Missing Values Before Cleaning (`df.isnull().sum()`)

Column	Missing Values
pakistan_national_avg_pm25	360
pakistan_global_rank	360

Missing values before cleaning.

Missing Values After Cleaning (0 across all 21 columns)

Column	Missing Values
year	0
month	0
month_name	0
city	0
province	0
region	0
latitude	0
longitude	0
population_millions	0
pm25_ugm3	0

Missing values after cleaning — 0 across all 21 columns.

Feature Analysis

The core numerical features analyzed are `pm25_ugm3` (fine particulate matter concentration), `aqi_us` (US Air Quality Index), `annual_pm25_avg_ugm3` (city's yearly baseline), `who_guideline_ratio` (multiple of the WHO safe limit), and `population_millions`. Across all 1,320 records, mean PM2.5 is 74.0 $\mu\text{g}/\text{m}^3$ (range: 11.7 – 266.3) and mean AQI is 143.2 (range: 49 – 317, i.e. spanning "Good" to "Hazardous"). Grouped comparisons show a clear geographic and seasonal skew rather than a uniform distribution:

Average AQI by City (Highest to Lowest)

City	Mean AQI	Min AQI	Max AQI	Median AQI
Faisalabad	190.1	108	317	173.5
Gujranwala	181.7	108	271	170.0
Multan	174.2	98	258	166.0
Lahore	170.8	97	293	166.5
Peshawar	136.0	72	194	149.0
Quetta	128.0	73	176	134.5
Hyderabad	118.9	63	177	117.0
Rawalpindi	117.7	57	176	109.0
Karachi	109.0	54	173	100.5
Islamabad	105.3	49	165	93.0

Average, minimum, maximum, and median AQI by city, ranked highest to lowest.

Faisalabad, Gujranwala, Multan, and Lahore — all in Punjab's industrial belt — form a distinct high-pollution cluster (mean AQI 170–190), while Islamabad and Karachi anchor the low-pollution end (mean AQI ~105–109) despite Karachi being the most populous city in the dataset (16.1 million people).

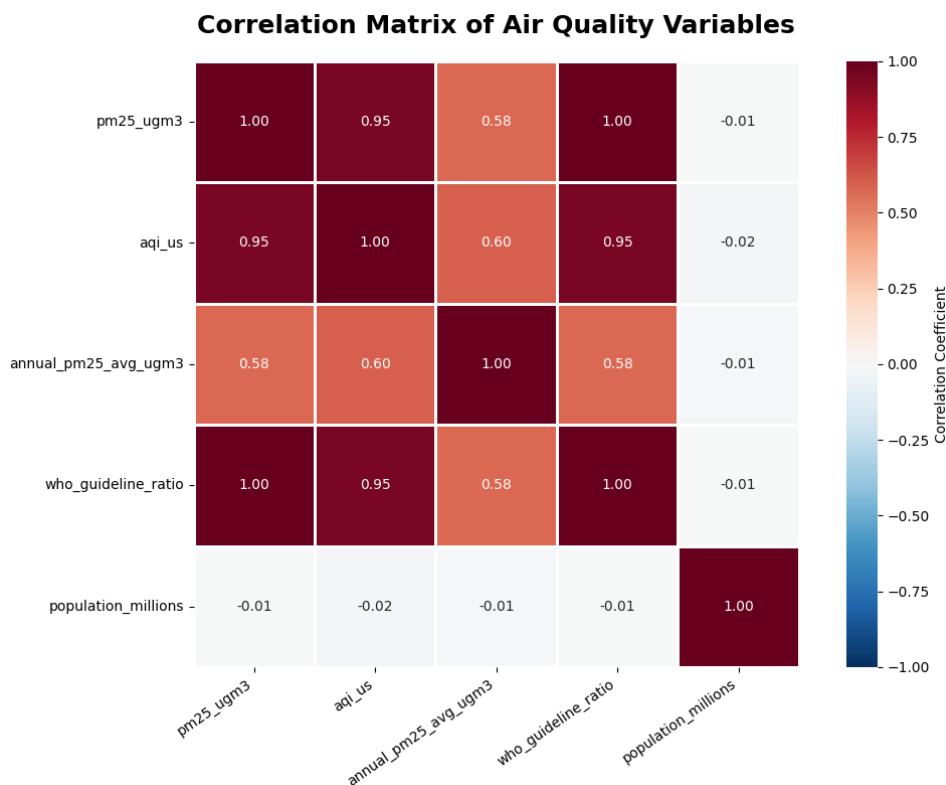
Average AQI by Season

Season	Average AQI
Winter (Smog Season)	196.9
Post-Monsoon	159.1
Spring	128.8
Summer/Monsoon	98.2

Average AQI by season — Winter (Smog Season) is roughly double Summer/Monsoon.

Correlation Analysis

A correlation matrix was computed across pm25_ugm3, aqi_us, annual_pm25_avg_ugm3, who_guideline_ratio, and population_millions.



Correlation heatmap of air quality variables.

Correlation Pairs — Strength Ranking

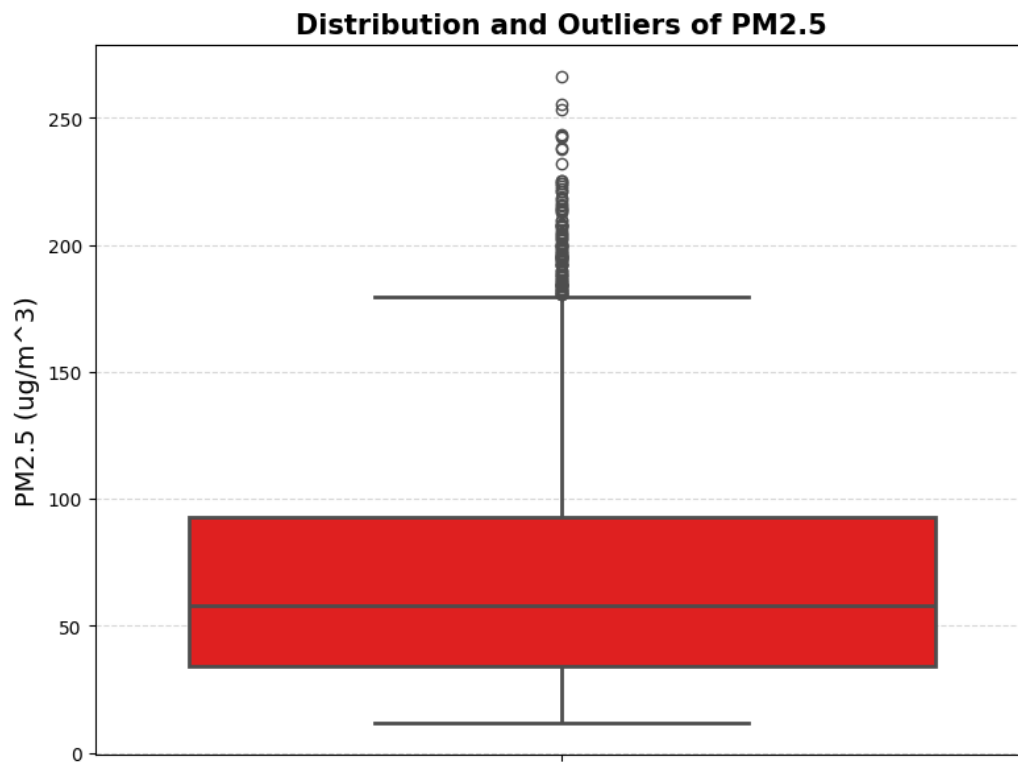
Feature 1	Feature 2	Correlation
pm25_ugm3	who_guideline_ratio	1.0
pm25_ugm3	aqi_us	0.952
aqi_us	who_guideline_ratio	0.952
aqi_us	annual_pm25_avg_ugm3	0.598
pm25_ugm3	annual_pm25_avg_ugm3	0.577
annual_pm25_avg_ugm3	who_guideline_ratio	0.577
pm25_ugm3	population_millions	-0.006
who_guideline_ratio	population_millions	-0.006
annual_pm25_avg_ugm3	population_millions	-0.008
aqi_us	population_millions	-0.019

Correlation pairs ranked by strength (self-pairs removed).

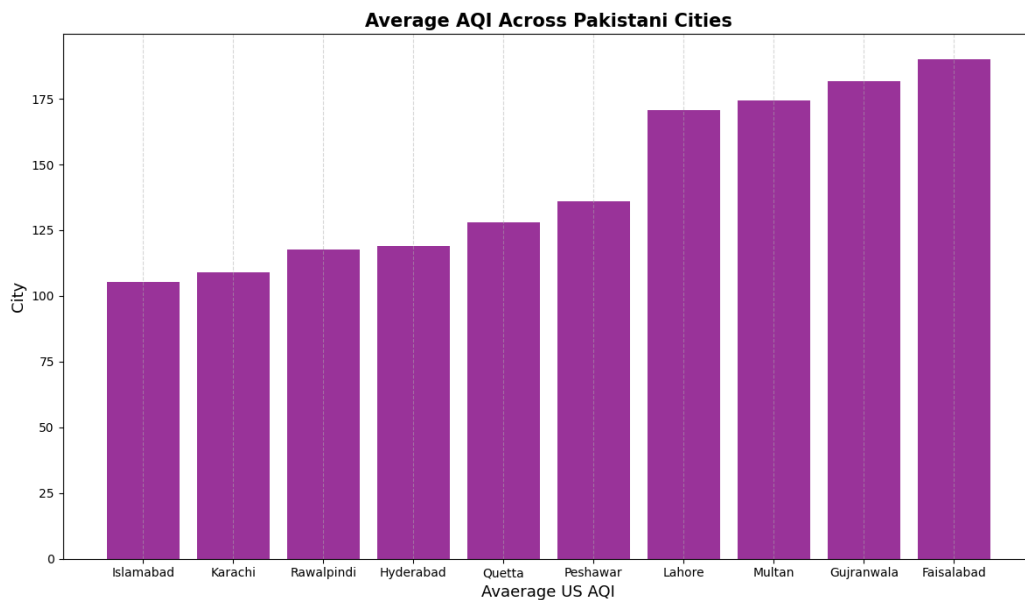
Strong positive relationships ($r \geq 0.70$): PM2.5 and the WHO guideline ratio are almost perfectly correlated ($r = 1.00$ — they are two scalings of the same measurement), and PM2.5 correlates very strongly with AQI itself ($r = 0.95$), confirming PM2.5 is the dominant pollutant driving Pakistan's AQI readings. **Moderate relationships ($0.30 \leq |r| < 0.70$):** AQI vs. annual city-level average ($r = 0.60$) and PM2.5 vs. annual average ($r = 0.58$) show that a city's yearly baseline meaningfully, but not perfectly, predicts any given month's reading. **No strong negative relationships** were found anywhere in the dataset. **Population size is uncorrelated with pollution** ($r \approx -0.02$ with both PM2.5 and AQI) — the clearest and most business-relevant finding of the correlation analysis, since it rules out population as a basis for prioritizing intervention spending.

Data Visualizations

The following 13 visualizations (exceeding the 12+ requirement) illustrate the distributional, geographic, seasonal, and correlational patterns uncovered during analysis.

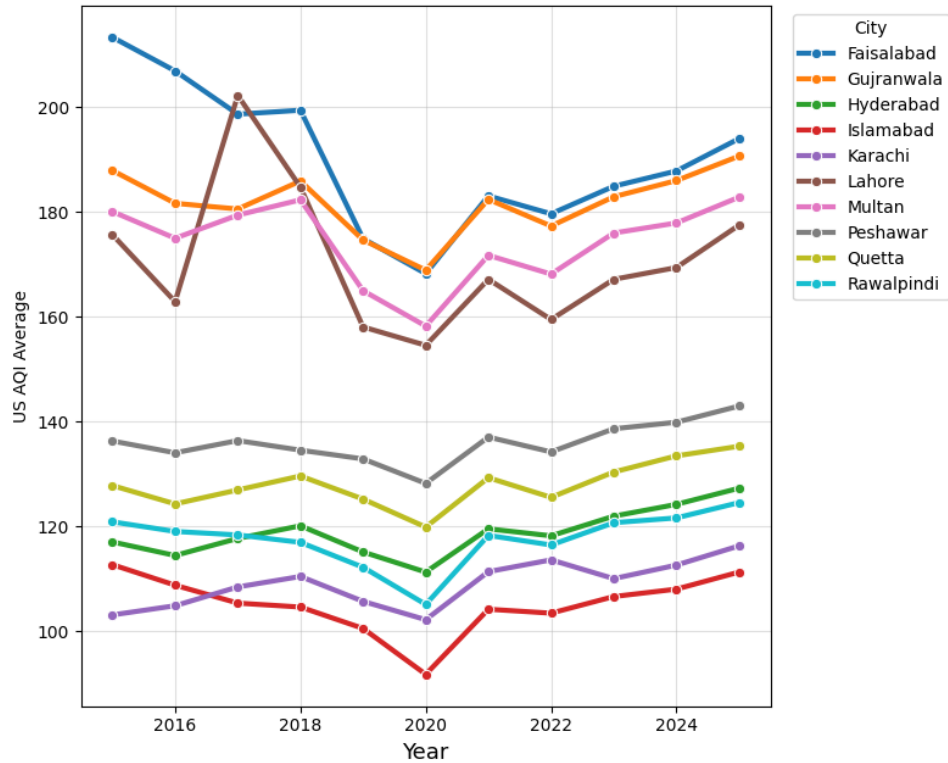


Boxplot of PM2.5 distribution — reveals the winter-month high-value outliers concentrated above the upper whisker.



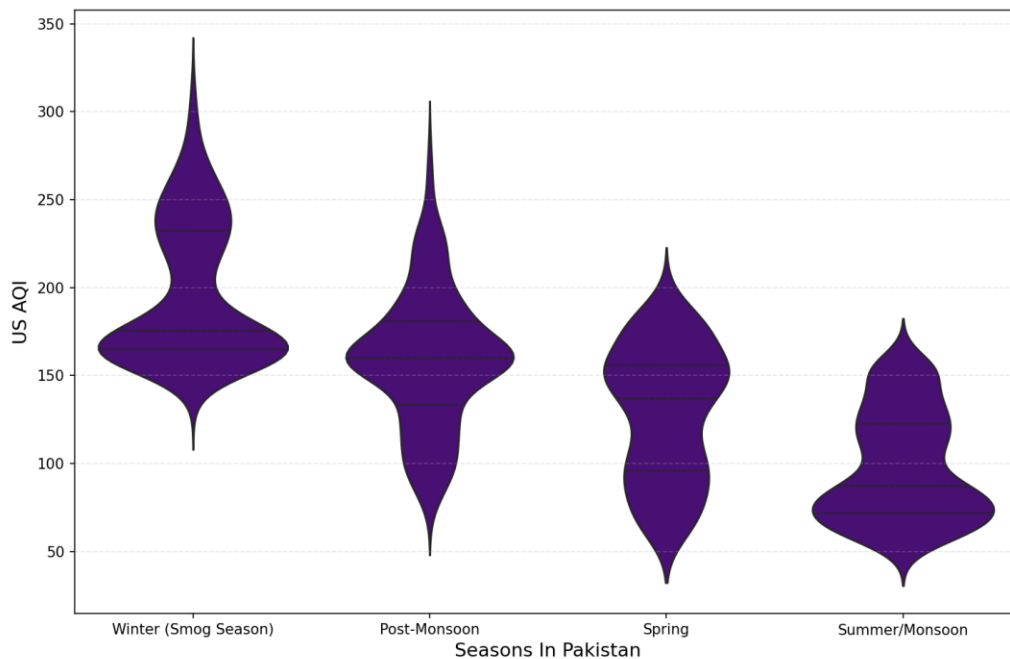
Average AQI by city, sorted lowest to highest — visualizes the ~2x gap between Islamabad and Faisalabad.

ÁQI yearly Trends Across Pakistani Cities (2015-2025)

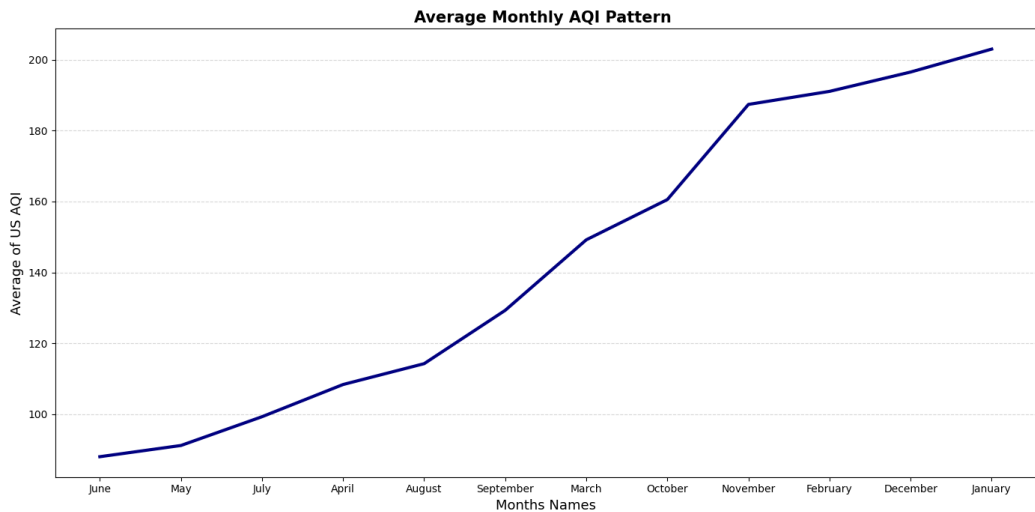


Year-over-year AQI trend per city (2015–2025) — shows the industrial Punjab cities consistently above the rest.

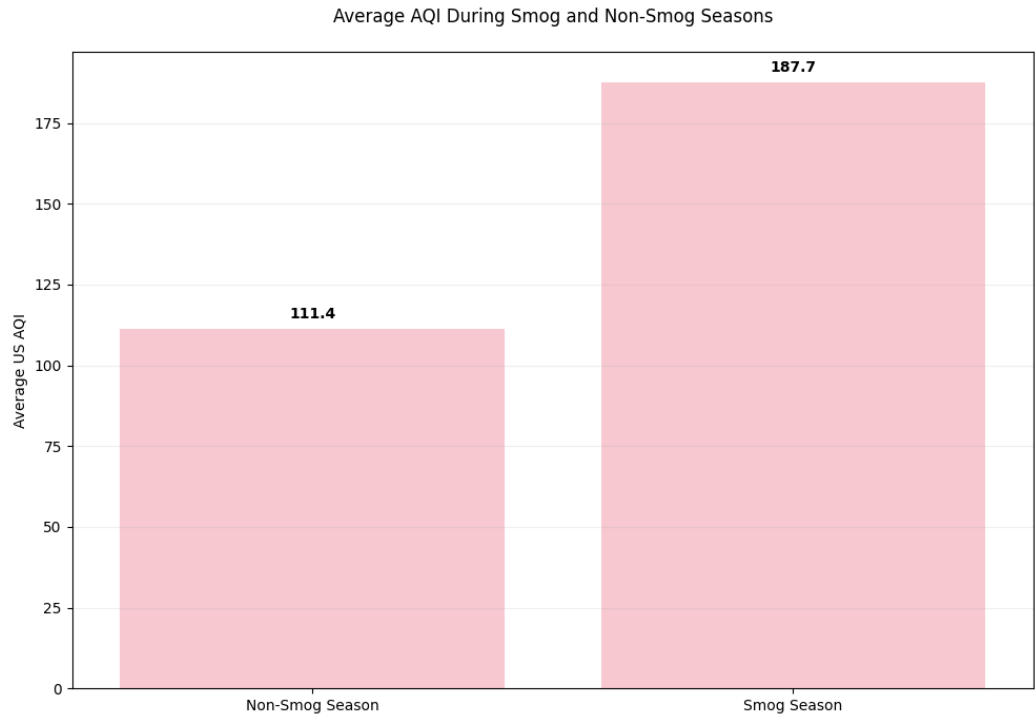
AQI Distribution Across Seasons



Violin plot of AQI distribution by season (season labels corrected to match the dataset) — Winter shows both the highest median and widest spread.

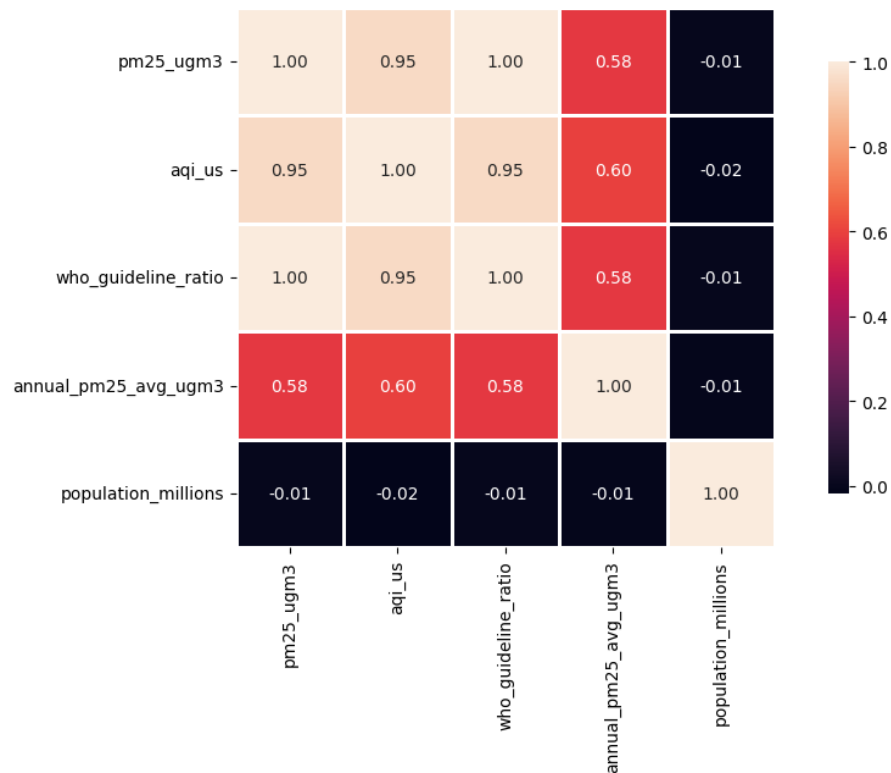


Average monthly AQI pattern across the full year — a clear winter peak / summer trough cycle.



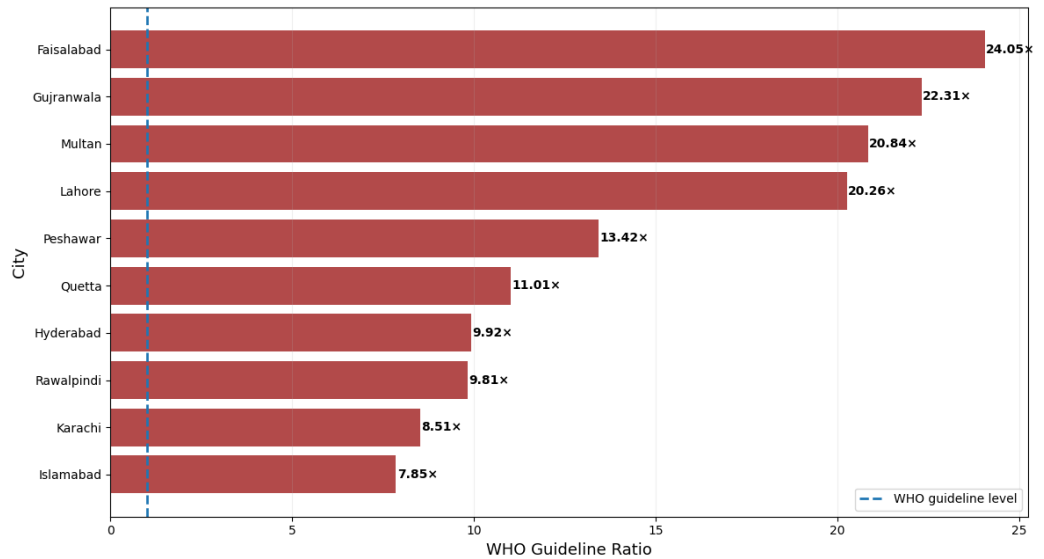
Average AQI: Smog season vs. non-smog season, with value labels.

Correlation Among Air Quality and Population Variables



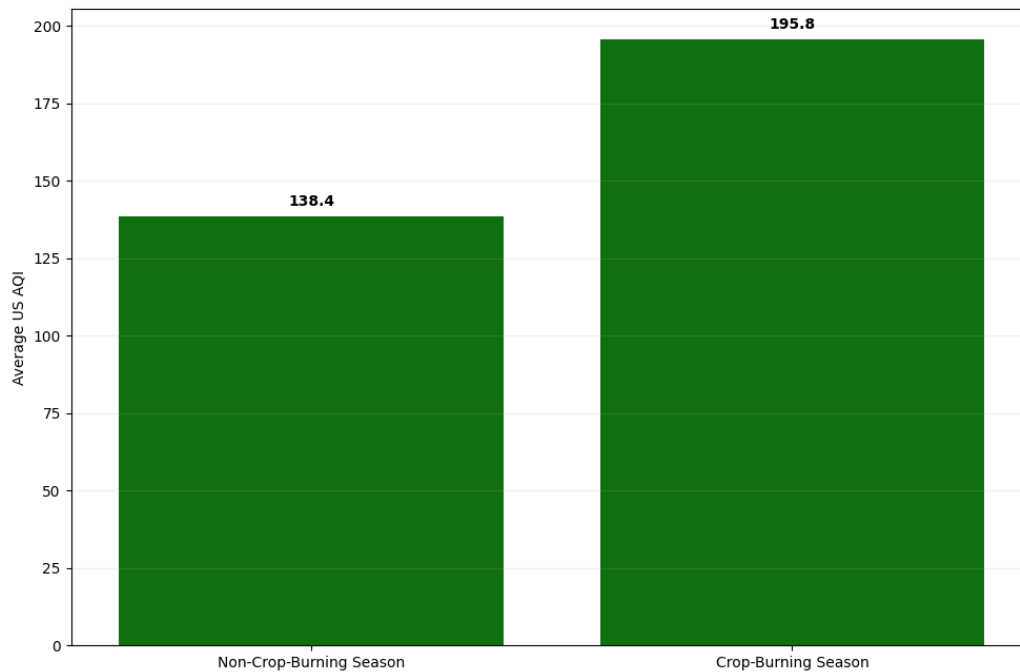
Correlation heatmap focused on pollution and population variables.

Average PM2.5 Level Relative to WHO Guideline



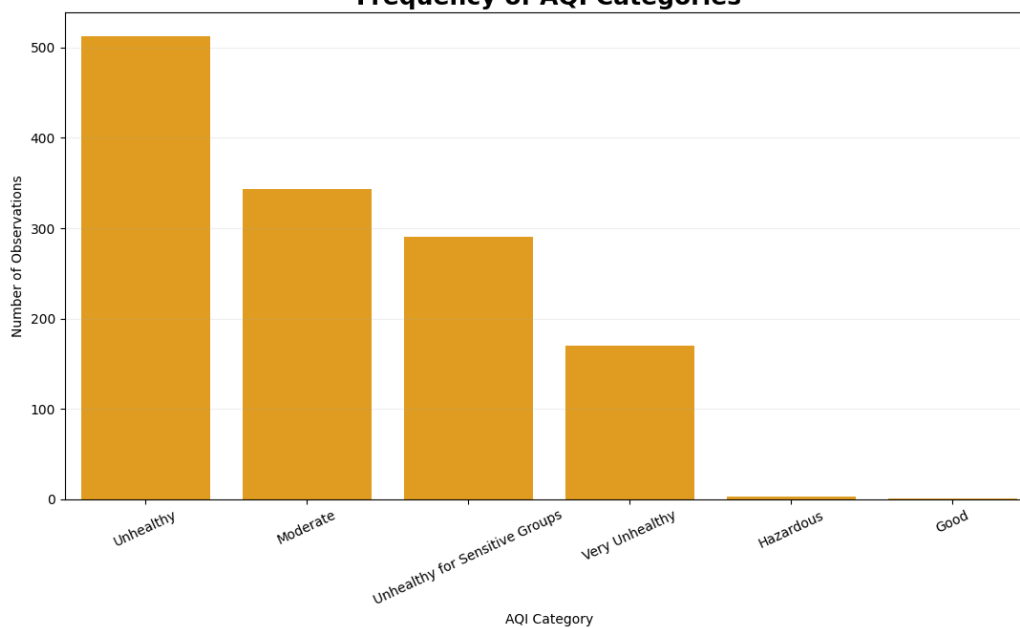
Average PM2.5 relative to the WHO guideline, by city (dashed line = WHO limit).

Average AQI During Crop-Burning and Non-Crop-Burning Seasons

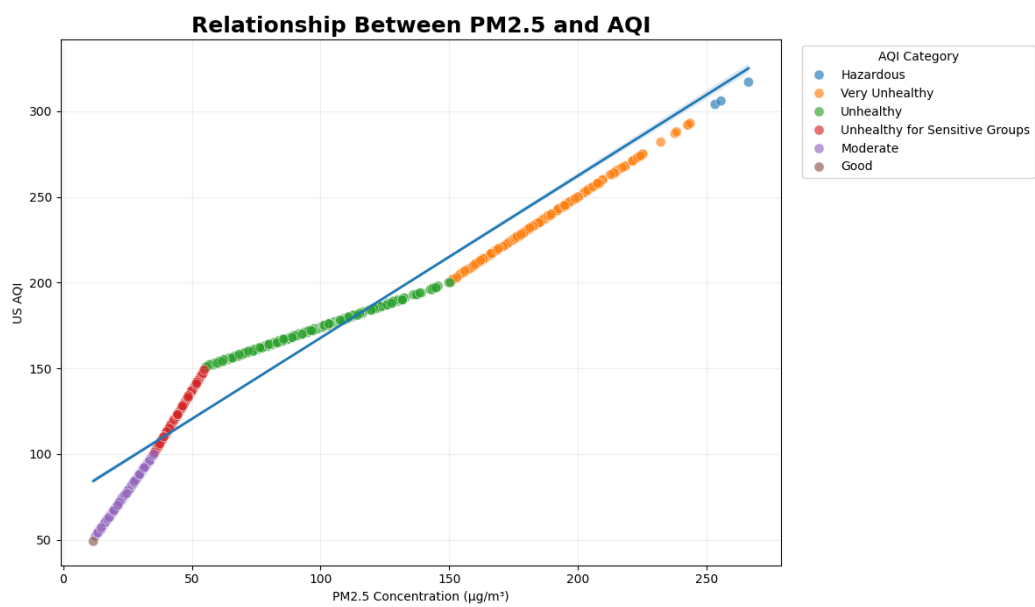


Average AQI: Crop-burning season vs. non-crop-burning season, with value labels.

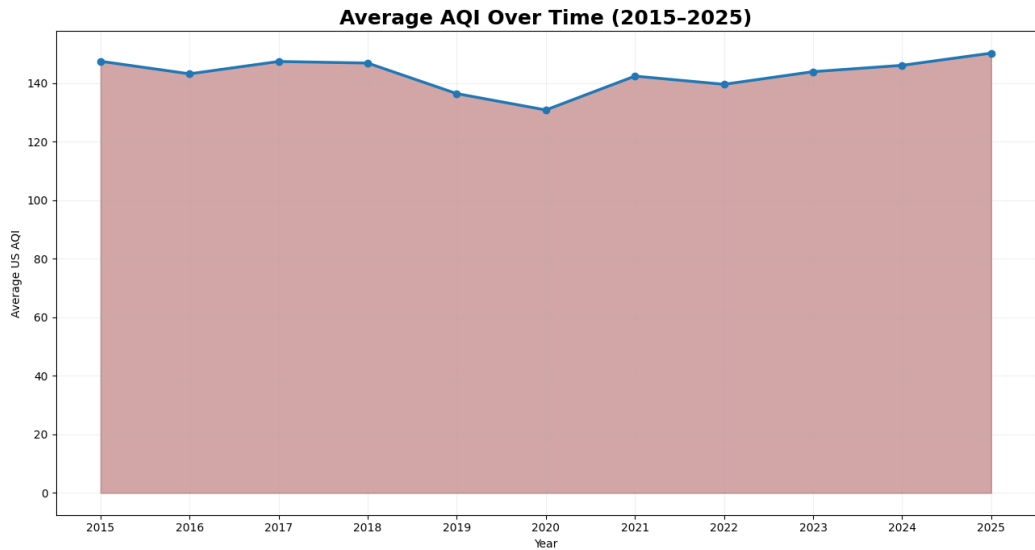
Frequency of AQI Categories



Frequency of each AQI category across all 1,320 observations.



Scatter plot of PM2.5 vs. AQI with regression line and AQI-category coloring.



Average AQI over time (2015–2025) — the COVID-era dip and subsequent rise to an all-time high in 2025.

Business Insights

The following 16 insights (exceeding the 15 required) are each grounded directly in the statistics and visualizations produced above.

- 1. Chronic, not occasional, pollution problem** — Across all 1,320 monthly records, the average AQI is 143.2 ("Unhealthy"). 52.0% of all observations fall in the Unhealthy/Very Unhealthy/Hazardous range, while only 26.1% are Good/Moderate — air quality is unsafe for roughly half of all city-months.
- 2. Industrial Punjab cities dominate pollution** — Faisalabad (190.1 avg AQI), Gujranwala (181.7), Multan (174.2), and Lahore (170.8) are the four most polluted cities, all in Punjab's industrial belt.
- 3. Population size does not drive pollution** — Correlation between population and AQI is effectively zero ($r = -0.019$). Karachi (16.1M) and Lahore (14.0M) are the largest cities, yet smaller industrial cities like Faisalabad (3.8M) are far more polluted.
- 4. Nearly 2x gap between best and worst cities** — Faisalabad's average AQI (190.1) is 1.81x Islamabad's (105.3), an 84.8-point gap between the most and least polluted cities.
- 5. Winter smog season roughly doubles AQI** — Winter (Smog Season) AQI averages 196.9 vs. 98.2 in Summer/Monsoon — almost exactly a 2x seasonal swing.
- 6. January is the worst month, June the best** — January averages 203.0 AQI vs. June's 88.0, a 2.3x difference. The three worst months (Jan, Dec, Feb) form a consistent winter cluster.
- 7. Crop-burning season adds a substantial penalty** — Average AQI during crop-burning months is 195.8 vs. 138.4 otherwise, a 41% jump, confirming agricultural burning materially worsens air quality.
- 8. COVID lockdowns proved activity reduction works** — During lockdown periods, average AQI fell to 98.5 vs. 144.6 outside lockdowns, a 32% reduction — a natural experiment showing cutting activity measurably improves air quality.
- 9. PM2.5 is the dominant driver of AQI** — PM2.5 and AQI correlate almost perfectly ($r = 0.952$), meaning fine particulate matter is what is driving Pakistan's poor AQI ratings.
- 10. Extreme WHO guideline exceedance in industrial cities** — Faisalabad's PM2.5 averages 24.0x the WHO guideline (peaking at 53.3x in winter); even Islamabad, the cleanest city, still averages 7.9x the WHO limit.
- 11. Pollution is rising again post-COVID, now at an all-time high** — National average PM2.5 dropped to 59.0 $\mu\text{g}/\text{m}^3$ in 2020 but climbed steadily to 80.1 $\mu\text{g}/\text{m}^3$ by 2025, its highest recorded level.
- 12. Most extreme outliers cluster in Faisalabad and Lahore winters** — The five worst monthly readings (PM2.5 up to 266.3 $\mu\text{g}/\text{m}^3$, AQI 317 "Hazardous") all occur in Faisalabad or Lahore during Nov–Jan.
- 13. Structural, mid-range pollution is the real issue, not rare disasters** — Only 1 "Good" and 3 "Hazardous" records exist out of 1,320. The dominant categories are "Unhealthy" (513) and "Unhealthy for Sensitive Groups" (290) — a chronic problem, not isolated catastrophes.
- 14. Monsoon rain measurably cleans the air** — AQI during monsoon months averages 114.9 vs. 151.4 in non-monsoon months, a ~24% reduction, confirming precipitation as a natural mitigating factor.
- 15. No natural factor other than rain/lockdowns reduces pollution** — Moderate correlations (0.58–0.60) exist between city-level annual averages and monthly readings, but no strong negative correlations appear anywhere — pollution only drops through deliberate intervention.
- 16. Provincial pattern reinforces the Punjab focus** — Punjab has by far the highest provincial average AQI (166.9) vs. Khyber Pakhtunkhwa (136.0), Balochistan (128.0), Sindh (113.9), and Islamabad Capital Territory (105.3).

Business Improvement Recommendations

The following 6 recommendations (exceeding the 5 required) are each directly tied to a finding above.

1. Concentrate intervention resources on the "Big Four" industrial Punjab cities (Faisalabad, Gujranwala, Multan, Lahore).

Why: These four cities consistently top the AQI rankings, and population size shows no correlation with pollution ($r = -0.019$), so budget allocation should follow the pollution data rather than population — targeting these four yields the highest health-impact return per resource spent.

2. Design and enforce time-boxed seasonal interventions for October–February rather than uniform year-round policy.

Why: Winter AQI (196.9) is roughly double Summer/Monsoon AQI (98.2), and crop-burning season alone adds a 41% AQI penalty. A synchronized push concentrated in this five-month window is far more cost-effective than spreading the same budget evenly across the year.

3. Pilot controlled activity-reduction measures (e.g., odd-even vehicle schemes, temporary industrial throttling) in the worst cities during peak smog weeks.

Why: The COVID lockdown period is a real-world proof of concept — AQI fell 32% ($144.6 \rightarrow 98.5$) when activity was reduced, showing a deliberately-scheduled, less disruptive intervention could achieve a measurable improvement.

4. Treat the 2020–2025 upward trend as an urgent renewal signal, not a stable baseline.

Why: National PM_{2.5} fell to $59.0 \mu\text{g}/\text{m}^3$ in 2020 but has climbed every year since to an all-time high of $80.1 \mu\text{g}/\text{m}^3$ in 2025. Without renewed enforcement, the pandemic-era improvement will keep eroding.

5. Target health and consumer-protection services (respiratory care, air-purifier retail, health insurance add-ons) specifically at Faisalabad, Gujranwala, Multan, and Lahore during November–February.

Why: This is exactly where and when WHO-guideline exceedance peaks (20x–24x on average, up to 53x), making it the most identifiable and time-bound target segment — a level of need cities like Islamabad and Karachi do not share.

6. Benchmark and replicate whatever policy/planning differences exist in Islamabad and Karachi, the two consistently lowest-AQI cities.

Why: Karachi has the largest population in the dataset (16.1M) yet ranks among the cleanest, proving high population and high pollution are not linked. Studying what these cities do differently could inform transferable interventions for the higher-AQI Punjab cities.

Conclusion

This analysis of 1,320 monthly air-quality records across 10 Pakistani cities (2015–2025) confirms that air pollution in Pakistan is a chronic, geographically-concentrated, and seasonally-driven problem rather than a series of isolated events. Industrial Punjab cities — Faisalabad, Gujranwala, Multan, and Lahore — bear a disproportionate share of the burden, while population size plays no meaningful role in determining pollution levels. Winter months, crop-burning season, and reduced human/industrial activity (as seen during COVID-19 lockdowns) all produce large, measurable swings in air quality, showing that targeted, seasonally-timed intervention can be highly effective. With national PM2.5 levels climbing back to an all-time high by 2025 after the pandemic-era dip, the window for renewed, geographically-targeted action is urgent. The 16 insights and 6 recommendations in this report provide a data-driven basis for prioritizing where, when, and how air-quality interventions and related business/health services should be deployed.

GitHub Repository Link

Project code and notebook:

https://github.com/MalaikaJabeen/Devixo_Solutions_Data_Science_Internship_projects/tree/main

Appendix — Python Code Screenshots

The screenshots below show representative Python code used at each stage of the analysis. Screenshots of the dataset, cleaning results, analysis-result tables, and visualizations appear inline within the corresponding sections above (Data Cleaning Process, Feature Analysis, Correlation Analysis, and Data Visualizations).

```
1 # Importing Libraries
2 import numpy as np
3 import pandas as pd
4 import matplotlib.pyplot as plt
5 import seaborn as sns
6
7 # Importing dataset
8 df = pd.read_csv("pakistan_air_quality_monthly_2015_2025.csv")
9 df.head()
```

Importing libraries and loading the dataset.

```
1 # Shape and size of dataset
2 df.shape           # (1320, 23)
3 df.size            # 30360
4
5 # Structural checks
6 df.info()
7 df.dtypes
8 df.columns
```

Initial structural inspection of the dataset.

```
1 # Data validation checks
2 df.duplicated().sum()           # 0
3 df['year'].min(), df['year'].max() # 2015, 2025
4 df['city'].value_counts()       # 132 records per city (balanced panel)
5 df['aqi_category'].value_counts()
```

Data validation checks after cleaning.

```
1 # Which city has the highest and lowest average AQI?
2 df.groupby('city')['aqi_us'].agg(['mean', 'min', 'max', 'median']) \
3   .sort_values('mean', ascending=False)
4
5 # Province and seasonal comparisons
6 df.groupby('province')['aqi_us'].mean().sort_values(ascending=False)
7 df.groupby('season')['aqi_us'].mean().sort_values(ascending=False)
```

Group-by analysis: AQI by city, province, and season.

```
1 # Seasonal-event impact on AQI
2 df.groupby("is_smog_season")["aqi_us"].mean()
3 df.groupby("is_crop_burning_season")["aqi_us"].mean()
4 df.groupby("covid_lockdown_period")["aqi_us"].mean()
```

Analyzing the impact of smog, crop-burning, and lockdown periods.

```
1 # Example visualization: PM2.5 vs AQI relationship
2 plt.figure(figsize=(12, 7))
3 sns.scatterplot(data=df, x="pm25_ugm3", y="aqi_us",
4                 hue="aqi_category", s=65, alpha=0.65)
5 sns.regplot(data=df, x="pm25_ugm3", y="aqi_us", scatter=False)
6 plt.title("Relationship Between PM2.5 and AQI", fontsize=18, fontweight="bold")
7 plt.tight_layout()
8 plt.show()
```

Example visualization code: PM2.5 vs. AQI scatter plot.